SC7 Bayes Methods MT25

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Lecture 3: Exchangeability and its implications

(updated to correct a typo and reorder slides as lecture)

Notation

If $X_1, ..., X_n$ are discrete with PMF $p_{1:n}(x_1, ..., x_n)$ on $\mathcal{X}^n$ then

$$P(X_1 = x_1, ..., X_n = x_n) = p_{1:n}(x_1, ..., x_n).$$

Subscript $1\!:\!n$ on $p_{1:n}$ gives variables and order so for $\sigma \in \mathcal{P}_{[n]}$,

$$P(X_{\sigma_1} = x_1, ..., X_{\sigma_n} = x_n) = p_\sigma(x_1, ..., x_n)$$

and

$$P(X_1 = x_{\sigma_1}, ..., X_n = x_{\sigma_n}) = p_{1:n}(x_{\sigma_1}, ..., x_{\sigma_n}).$$

If $x = (x_1, \ldots, x_n)$ and

$$P(X_1 = x_1, ..., X_n = x_n) = P(Y_1 = x_1, ..., Y_n = x_n)$$

for all $x \in \mathcal{X}^n$ then we write

$$(X_1, ..., X_n) \sim (Y_1, ..., Y_n)$$

so $(X_1, ..., X_n)$ and $(Y_1, ..., Y_n)$ are “the same random variables”.

Marginal Consistency (preamble for exchangeability)

Let $[n] = \{1, \ldots, n\}$ and $\mathcal{S}_{[n]}$ be the set of all subsets of $[n]$ excluding $\emptyset$ and $[n]$ itself.

Let $r \in \mathcal{S}_{[n]}$ be a subset of $[n]$ with m elements, let $s = [n] \setminus r$ and let $p_s : \mathcal{X}^{n-m} \to [0, 1]$ be a PMF.

The family $\mathcal{F} = \{p_s,\ s \in S_{[n]}\}$ of PMFs is marginally consistent if the marginals

$$q_s(x_s) = \sum_{x_{r_1} \in \mathcal{X}} \cdots \sum_{x_{r_m} \in \mathcal{X}} p_{1:n}(x_1, \ldots, x_n)$$

satisfy $q_s(x_s) = p_s(x_s)$ for all $s \in \mathcal{F}$ and $x_s \in \mathcal{X}^{n-m}$.

The family is closed under taking marginals (AKA projective).

When do we *not want* marginal consistency in our prior models?

Example: students with the same birthday as Bruno de Finetti and students asking a question in a lecture.

Class with students $\{1, \ldots, n\}$ and class with just student $\{1\}$.

Let $X = (X_1, \ldots, X_n)$ with $X_i \in \{0, 1\}$ and $X \sim p_{1:n}$.

Let $Y \in \{0, 1\}$ with $Y \sim p_1$.

If $X_i = 1/0$ if student $i \in [n]$ has/not birthday on 13th June then we want $Y \sim X_1$ in our priors for the two classes (ie MC with $r = 2 : n$ and $s = 1$).

If $X_i = 1/0$ if student $i \in [n]$ asks/doesn't a question in class then requiring $Y \sim X_1$ in our priors for the two classes doesn't match my experience with students!

The key idea here is that when we model parameters with a prior, and we write down a family of priors $p_s,\ s \in \mathcal{F}$, we should be aware if we are violating MC.

Marginal Consistency (examples)

Let $x \in \{0,1\}^n$ and $k(x) = \sum_{i=1}^n x_i$. The distributions

$$p_{1:n}(x) = c_n 2^{-k(x)(n-k(x))}, \qquad (1)$$

(with $c_3 = 2/7$ and $c_4 = 8/27$) are not marginally consistent as

$$p_{1:3}(0,0,0) \neq p_{1:4}(0,0,0,0) + p_{1:4}(0,0,0,1)$$

as $p_{1:3}(0,0,0) = 2/7$ and $p_{1:4}(0,0,0,0) + p_{1:4}(0,0,0,1) = 1/3$.

By contrast

$$p_{1:n}(x) = \frac{k(x)!(n-k(x))!}{(n+1)!}, \quad n \geq 1 \qquad (2)$$

is marginally consistent. If $\theta \sim U(0,1)$ and $X_i \sim \mathrm{Bern}(\theta)$ iid for $i = 1, \ldots, n$ then

$$p_{1:n}(x) = \int_0^1 \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} d\theta. \qquad (3)$$

and we can knock out any element: $\sum_{x_1} p_{1:n}(x) = p_{2:n}(x_2, \ldots, x_n)$.

Infinite Exchangeable Sequence (IES)

Random variables $\{X_i\}_{i=1}^{\infty}$ are an IES if

$$(X_1, X_2, ..., X_n) \sim (X_{\sigma_1}, X_{\sigma_2}, ..., X_{\sigma_n})$$

for each $n \geq 1$ and all $\sigma \in \mathcal{P}_{[n]}$. Equivalently,

$$p_{1:n}(x_1, \ldots, x_n) = p_{1:n}(x_{\sigma_1}, \ldots, x_{\sigma_n})$$

for all $x \in \mathcal{X}^n$ and $\sigma \in \mathcal{P}_{[n]}$.

Example: a conditionally independent sequence of binary random variables $\theta \sim \pi(\cdot)$, $X_i \sim \text{Bern}(\theta)$, $i = 1, ..., n$ iid given θ is exchangeable as

$$p(x_1, ..., x_n) = \int_0^1 \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}\pi(\theta)d\theta.$$

Permuting indices just shuffles the product.

Question: Taking $\theta \sim U(0,1)$ we get (3) so there is an IES with marginals in (2). Does an IES with marginals given by (1) exist?

Theorem (de Finetti): Let $X_1, X_2, \dots X_n, \dots$ be an infinite sequence of binary random variables. The sequence is exchangeable if and only if there exists a distribution function $F(\theta)$ on $[0,1]$ such that

$$p(x_1, \dots, x_n) = \int_0^1 \left[\prod_{i=1}^{n} p(x_i|\theta) \right] dF(\theta) \qquad (*)$$

with

$$F(\theta) = \Pr(\Theta \leq \theta) \quad \text{where} \quad \Theta = \lim_{N \to \infty} N^{-1} \sum_{i=1}^{N} X_i.$$

It further holds that the conditioned distribution is Bernoulli,

$$p(x_1, \dots, x_n | \Theta = \theta) = \prod_{i=1}^{n} p(x_i|\theta),$$

with $p(x_i|\theta) = \theta^{x_i}(1-\theta)^{1-x_i}$.

Remarks $\{X_i\}_{i=1}^{\infty}$ if an IES iff a generative model

$$\Theta \sim F, \quad X_i \sim p(\cdot|\theta), \text{ iid for } i = 1, 2, 3, ... \text{ when } \Theta = \theta.$$

exists for some F and $p(\cdot|\theta)$. See Bernoulli example in (3).

The theorem extends to cover infinite exchangeable sequences of random vectors (ie X_i continuous multivariate random variables) with a multivariate parameter θ.

The expression $dF(\theta)$ may be off-putting. For example, if $F(\theta)$ is the cdf of a pdf π then $dF(\theta) = \pi(\theta)d\theta$, and

$$p(x_1, ..., x_n) = \int_0^1 \prod_{i=1}^{n} p(x_i|\theta)\pi(\theta)d\theta.$$

In general F is just some unknown distribution which puts probability mass $F(A)$ in some sigma-field of subsets of Ω.

Outline proof (of deF): $IES \Leftarrow (*)$ from the permutation symmetry in $(*)$. For $IES \Rightarrow (*)$ we will take n of the first N outcomes in the sequence, write down the joint pmf of $X_1, ..., X_n$ and show that it converges to the RHS of Equation $(*)$ as $N \to \infty$. Let

$$S_n = X_1 + X_2 ... + X_n, \quad n = 1, 2, ...$$

and let r, s be two integers $0 \le r \le s \le N$. By exchangeability, the distribution of S_n given $S_N = s$ is hypergeometric,

$$\Pr(S_n = r | S_N = s) = \frac{\binom{s}{r}\binom{N-s}{n-r}}{\binom{N}{n}}$$

since this is the probability to draw r 1's in n draws without replacement from an urn containing s 1's and $N - s$ 0's.

When $S_n = r$ the urn contains at least $n - r$ 0's so

$$\begin{aligned}
\Pr(S_n = r) &= \sum_{s=r}^{N-(n-r)} \Pr(S_n = r | S_N = s) \Pr(S_N = s) \\
&= \sum_{s=r}^{N-(n-r)} \Pr(S_n = r | S_N/N = \theta(s)) \Pr(S_N/N = \theta(s))
\end{aligned}$$

where $\theta(s) \equiv s/N$.

The random variable

$$\Theta_N = S_N/N$$

takes values $\Theta_N \in \{0, 1/N, 2/N, ..., 1\}$. It has CDF F_N with

$$\begin{aligned}
F_N(\theta) &= \Pr(\Theta_N \le \theta), \\
&= \sum_{s=0}^{N} \Pr(S_N/N = \theta(s)) \mathbb{I}_{\theta(s) \le \theta}
\end{aligned}$$

Now "differentiate the step function"

$$dF_N(\theta) = \sum_{s=0}^{N} \Pr(S_N = N\theta(s))\delta_{\theta(s)}(d\theta)$$

and write $\Pr(S_n = r)$ as an integral

$$\begin{aligned}\Pr(S_n = r) &= \sum_{s=r}^{N-(n-r)} \Pr(S_n = r | S_N/N = \theta(s)) \Pr(S_N/N = \theta(s)) \\ &= \int_{r/N}^{1-(n-r)/N} \Pr(S_n = r | S_N = N\theta) dF_N(\theta).\end{aligned}$$

as $r \leq s \leq N-(n-r)$ is the same as $r/N \leq \theta \leq 1-(n-r)/N$.

Exercise: very the last step by substituting in dF_N and doing the integral (integrating δ-functions is easy as you just substitute $\theta(s)$ for θ in the integrand).

We now assume two limiting results. The hypergeometric $\Pr(S_n = r|S_N = N\theta)$ converges uniformly to the binomial,

$$\frac{\binom{N\theta}{r}\binom{N(1-\theta)}{n-r}}{\binom{N}{n}} \to \binom{n}{r}\theta^r(1-\theta)^{n-r}$$

as $N \to \infty$ at fixed θ. Sampling a small number from a large population with or without replacement is the same.

Helly's Theorem (Feller (1966) *Probability Theory and Applications* II) "an infinite sequence of probability distributions F_N on a finite interval contains a convergent subsequence".

It follows that a limit $F_N(\theta) \to F(\theta)$ (pointwise) exists, that is

$$\lim_{N\to\infty} \Pr(\Theta_N \le \theta) = \Pr(\Theta \le \theta)$$

with $\Theta \sim F$.

Assuming these two convergence results,

$$\begin{aligned}\Pr(S_n = r) &= \lim_{N\to\infty} \int_{r/N}^{1-(n-r)/N} \Pr(S_n = r \mid S_N = N\theta)\, dF_N(\theta) \\ &= \binom{n}{r} \int_0^1 \theta^r (1-\theta)^{n-r} dF(\theta) \qquad (A)\end{aligned}$$

By exchangeability of $X_1, ..., X_n$,

$$\Pr(S_n = r) = \binom{n}{r} p(x_1, ..., x_n) \qquad (B)$$

for any $x_1, ..., x_n$ summing to r, so equating (A) and (B),

$$p(x_1, ..., x_n) = \int_0^1 \theta^r (1-\theta)^{n-r} dF(\theta) \qquad (C)$$

which is the result of de Finetti.

The representation in terms of F is unique. A distribution on a bounded interval is uniquely determined by its moments, and (C) fixes all the moments $E(\theta^n), n = 1, 2, ...$ of F (take $r = n$ and $n = 1, 2, ...$) so F exists and is unique. [EOP]

Bayesian inference (existence of prior)

If the data $X_1 = x_1, ..., X_n = x_n$ come from an IES then

$$p(x_1, ..., x_n|\theta) = \prod_i p(x_i|\theta)$$

is the likelihood. The generative model is

$$\Theta \sim F \quad \text{and} \quad X_i \sim p(\cdot|\theta),\ i = 1, ..., n$$

so a "true" prior F exists.

We cant often elicit the true generative model. We do our best to write down π to match F. At least the random variable Θ we are trying to model exists.

A similar issue arises for the likelihood - we know a conditional distribution for $X|\theta$ exists and do our best to approximate it.

Bayesian inference: (posterior)

Suppose we have seen $x_1, ..., x_m$ and we wish to predict $x_{m+1}, ..., x_n$. The predictive distribution is

$$\begin{aligned} p(x_{m+1:n}|x_{1:m}) &= p(x_{1:n})/p(x_{1:m}) \\ &= \int p(x_{m+1:n}|\theta)\frac{p(x_{1:m}|\theta)dF(\theta)}{p(x_{1:m})}. \end{aligned}$$

We see that

$$dF(\theta|x_1, ..., x_m) \propto p(x_1, ..., x_n|\theta)dF(\theta)$$

is our updated prior for θ given $x_{1:m}$, ie, the posterior.

de Finetti tells us an "unknown true" posterior distribution exists.